Cantera vs NN | NRMSE summary [%]

Median of per-IC NRMSE on the same 15 complete test ICs for every model

$$\mathrm{NRMSE}(\%) = 100\,\mathrm{RMSE}_{time}/(\max_t y_{Cantera} - \min_t y_{Cantera})$$

Model / same-state mean	Training	T [K]	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
1 ANN10	**10k**	10.9	6.72	19.9	16	7.77	66.6	6.61	9.57	15.1	N/A	18/18
2 P-only DeepONet10	**10k**	20.7	8.69	13.5	12	2.76	44	15.7	5.65	14.6	N/A	18/18
2 P-only DeepONet10	**long**	29.1	25.1	5.2	19.8	8.19	19.5	29.2	1.22	4.64	N/A	18/18
3 ANN10 + P-only DeepONet10	**10k**	13.9	4.56	7.83	11	7.31	41.9	12.4	6.96	14.6	N/A	18/18
3 ANN10 + P-only DeepONet10	**long**	4.02	5.58	5.13	5.37	1.72	14.6	3.69	1.25	4.95	N/A	18/18
4 T/P DeepONet10	**10k**	19.7	5.81	15.5	9.45	5.21	58	10.7	10.8	16.7	N/A	17/18
4 T/P DeepONet10	**long**	60.6	15.3	87.9	6.73	2.68	230	25.9	4.62	14.1	N/A	16/18
5 ANN10 + T/P DeepONet10	**10k**	11.6	2.98	12	6.73	3.36	47.9	6.49	6.42	7.07	N/A	18/18
5 ANN10 + T/P DeepONet10	**long**	47.8	7.85	46.5	8.19	3.68	341	28.6	9.5	7.42	N/A	18/18
6 T/P/ϕ DeepONet10	**10k**	21	4.14	13.2	8.44	3.48	62.9	13.4	8.09	14.6	N/A	17/18
6 T/P/ϕ DeepONet10	**long**	42.2	10.8	42.2	6.33	0.94	158	23.3	9.58	9.3	N/A	18/18
7 ANN10 + T/P/ϕ DeepONet10	**10k**	27	3.21	6.12	3.19	3.34	64.4	13.8	5.99	8.02	N/A	18/18
7 ANN10 + T/P/ϕ DeepONet10	**long**	33.9	4.02	30	5.99	0.81	147	16.4	2.87	6.42	N/A	18/18

RMSE uses trapezoidal time integration on the common 2 us grid. The Cantera range uses the entire 10 ms reference and never NN values.
N/A: constant or nearly constant Cantera reference (N2 is constant here). —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
Common ICs: 4, 5, 7, 32, 39, 44, 45, 57, 64, 68, 69, 73, 74, 76, 88.